Grade 11 – Term 1
C1–C7 Catch-up Exam Version 2 Solutions
Continuous distributions and sample data

Global Economics – 2026–2027

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Learning objective

Identify probability distributions and their supports; use geometric interval probabilities and exponential waiting times; calculate theoretical moments, empirical probabilities, and sample statistics; and interpret errors relative to theoretical models in economic and operational contexts.

Evaluated criteria

C1. Characterises discrete and continuous random variables, identifies suitable distributions and supports, and interprets density and cumulative probability in context.

C2. Applies uniform and triangular models and constant or piecewise-linear densities to interval probabilities using geometric areas.

C3. Uses circular geometry and arc proportions to calculate probabilities for uniform timing and directions around a circle.

C4. Applies exponential models to constant-rate waiting times, infers event rates, and assesses model suitability.

C5. Calculates and interprets the mean, variance, and standard deviation of continuous distributions using standard formulas.

C6. Estimates interval probabilities using sample relative frequencies and compares them with theoretical probabilities.

C7. Calculates sample means, variances, and standard deviations for different sample sizes and data representations and compares them with theory.

Assessment information

Each task shows its points. Passing requires **at least 5 cumulative points per criterion, including previous learning activities**. Show all procedures; answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Missing names or late submissions incur a 1-point deduction per criterion.

Criterion	C1	C2	C3	C4	C5	C6	C7	Total
Points	6	6	6	7	6	6	6	43

Conventions. $N(\mu, \sigma^2)$ uses mean and variance; $\text{Tri}(a, b, c)$ means minimum, maximum and mode; $\text{Exp}(\lambda)$ uses the rate parameter. Supports include their finite boundary points; endpoint inclusion does not affect continuous probabilities. Probabilities are rounded to four decimal places (percentages to two), statistics, rates and times to three where needed, and percentage errors to two. All errors use unrounded results. Sample variance uses $n - 1$. Blue task text links to its solution; each solution links back to its task. The arrows return to the problem and contents.

1 Models in Colombian operations [C1-6]

Colombian organisations model two random variables in each context below. **For each variable**: classify it as discrete or continuous, and identify its distribution.

Questions/Tasks

C1 1. Municipal compost participation. A municipal programme records (a) whether one randomly selected household sorts organic waste correctly, coded 1 if correct and 0 otherwise; and (b) the number sorting correctly among 35 households. Each household sorts correctly independently with probability 0.86. **[1]**

C1 2. Water-meter auditing. A utility records (a) the number of meters tested until the first faulty meter, including that meter, with independent tests and constant fault probability 0.06; and (b) a calibration error between -2 and 4 litres, with equal-length intervals having equal probabilities and no errors outside these limits. **[1]**

C1 3. Cultural-centre bookings. A centre records (a) the number of rehearsal bookings received in one hour, arriving independently at a constant rate of 3 per hour, one at a time, with multiple arrivals in a sufficiently short interval negligible; and (b) a room-reset time on $[8, 26]$ minutes whose density rises linearly from zero at 8 to a single peak at 15, then falls linearly to zero at 26, with no values outside this range. **[1]**

C1 4. Solar-energy monitoring. A solar cooperative records (a) the usable fraction of a battery bank at sunset on $[0.2, 0.9]$, whose density increases along one straight line from zero at 0.2 to its maximum at 0.9 and is zero elsewhere; and (b) a sensor error modelled by an exact symmetric bell curve on the real line, centred at 1 watt with standard deviation 3 watts. **[1]**

C1 5. Agricultural training. A training service records (a) the number of cancellations among 45 independently booked places, each cancelled with probability 0.07; and (b) the number of registration calls over four days, arriving independently at a constant rate of 6 per day, one at a time, with multiple arrivals in a sufficiently short interval negligible. **[1]**

C1 6. Community-kitchen planning. A community kitchen studies (a) the time to prepare a vegetable batch, ranging from 12 to 36 minutes. Short preparation times are most common, and the likelihood of longer times decreases steadily to zero at the upper limit. (b) Daily rice use is concentrated near 18 kg, with lighter and heavier usage progressively less common in a symmetric pattern. Its standard deviation is 2.5 kg. **[1]**

Solution: Compost participation

(a) Discrete: $X \sim \text{Bernoulli}(0.86)$, support $\{0, 1\}$. One household gives a single correct/incorrect outcome. **(b)** Discrete: $Y \sim \text{Bin}(35, 0.86)$, support $\{0, 1, \ldots, 35\}$. This counts successes in 35 independent trials with the same probability.

Return to this task

Solution: Water-meter auditing

(a) Discrete: $X \sim \text{Geom}(0.06)$, support $\{1, 2, \ldots\}$. Count every test up to and including the first faulty meter; $P(X = k) = 0.94^{k-1}(0.06)$. **(b)** Continuous: $Y \sim U(-2, 4)$ litres, support $[-2, 4]$ L. Equal-length intervals have equal probabilities, so $f(y) = 1/(4 - (-2)) = 1/6$ L^{-1} on the support and zero elsewhere.

Return to this task

Solution: Cultural-centre bookings

(a) Discrete: $X \sim \text{Poisson}(3)$, support $\{0, 1, 2, \ldots\}$. The one-hour mean is $3 \times 1 = 3$ bookings; the stated conditions define a Poisson process. **(b)** Continuous: $Y \sim \text{Tri}(8, 26, 15)$ minutes, support $[8, 26]$ min. The straight branches specify a triangular density with mode 15.

Return to this task

Solution: Solar-energy monitoring

(a) Continuous: increasing-linear density, equivalently $X \sim \text{Tri}(0.2, 0.9, 0.9)$, support $[0.2, 0.9]$. Normalisation gives height $2/0.7 = 20/7$ and $f(x) = 2(x - 0.2)/(0.7)^2 = 200(x - 0.2)/49$ on this support, zero elsewhere. The fraction is unitless; its mode is 0.9. **(b)** Continuous: $Y \sim N(1, 9)$ in watts, support $\mathbb{R}$. The exact symmetric bell curve identifies normality, with SD 3 W and variance $3^2 = 9$ W^2.

Return to this task

Solution: Agricultural training

(a) Discrete: $X \sim \text{Bin}(45, 0.07)$, support $\{0, 1, \ldots, 45\}$. These are 45 independent cancellation trials with constant probability. **(b)** Discrete: $Y \sim \text{Poisson}(24)$, support $\{0, 1, 2, \ldots\}$. The four-day mean is $6 \times 4 = 24$ calls; 6 is a daily rate, not the four-day count parameter.

Return to this task

Solution: Community-kitchen planning

(a) Continuous, support $[12, 36]$ min. "Decreases steadily" does not uniquely establish a straight-line density. *If a decreasing straight line is intended*, use $X \sim \text{Tri}(12, 36, 12)$ with $f(x) = (36 - x)/288$ on this support and zero elsewhere. Its height is $2/24 = 1/12$ min^{-1}; no extra parameter is needed under this assumption. **(b)** Continuous rice use. Symmetric concentration near 18 kg and SD 2.5 kg do not uniquely determine normality or full support. *If normality is assumed*, $Y \sim N(18, 6.25)$ in kg, model support $\mathbb{R}$, with variance $2.5^2 = 6.25$ kg^2. Physical rice use is nonnegative, but its exact physical support is not supplied. These are candidate models under explicit assumptions.

Return to this task

2 Seedling-tray preparation [C2-6]

At a nursery in Boyacá, the time S to fill and seed a tray ranges from 4 to 28 minutes. Its density is zero at both endpoints, rises linearly to its single maximum at 14 minutes, then falls linearly to zero; it is zero outside $[4, 28]$.

Questions/Tasks

C2 Find $P(8 \le S \le 22)$. **[3]**

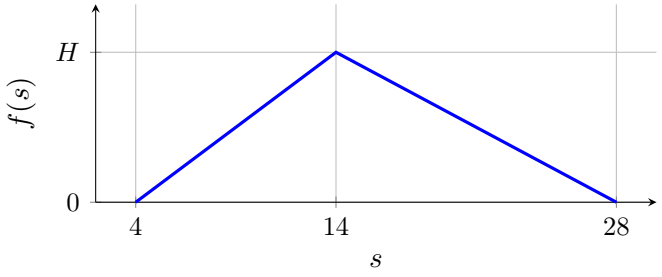
C2 Suppose the support changes to $[2, 34]$. Keep the triangular shape, the mode at 14 minutes, and the event bounds 8 and 22 minutes fixed. Recalculate $P(8 \le S \le 22)$. **[3]**

Solution: Original range

Stage 1: Support and shape

$S \sim \text{Tri}(4, 28, 14)$ has support $[4, 28]$ min. The density rises from $(4, 0)$ to $(14, H)$ and falls to $(28, 0)$.

Support and triangular shape; horizontal coordinate in minutes



Stage 2: Total area = 1

The triangle area is $\frac{1}{2} \times \text{base} \times \text{height}$, so

$$1 = \frac{1}{2}(28 - 4)H \quad \Longrightarrow \quad H = \frac{2}{24} = \frac{1}{12} \text{ min}^{-1}.$$

Stage 3: Complete piecewise pdf

Use $m = (y_2 - y_1)/(x_2 - x_1)$ and $q = y_1 - mx_1$ in $f(s) = ms + q$. For the rising line through $(4, 0)$ and $(14, \frac{1}{12})$,

$$m_1 = \frac{\frac{1}{12} - 0}{14 - 4} = \frac{1}{120}, \qquad q_1 = 0 - (\frac{1}{120})(4) = \frac{-1}{30}, \qquad f_1(s) = \frac{s - 4}{120}.$$

For the falling line through $(14, \frac{1}{12})$ and $(28, 0)$,

$$m_2 = \frac{0 - \frac{1}{12}}{28 - 14} = \frac{-1}{168}, \qquad q_2 = 0 - (\frac{-1}{168})(28) = \frac{1}{6}.$$

Thus the complete density is

$$f(s) = \begin{cases} \dfrac{s - 4}{120}, & 4 \le s \le 14, \\ \dfrac{28 - s}{168}, & 14 < s \le 28, \\ 0, & \text{otherwise.} \end{cases}$$

The branches meet at height $\frac{1}{12}$, are nonnegative, and enclose total area 1.

Stage 4: Probability by complementary tail triangles

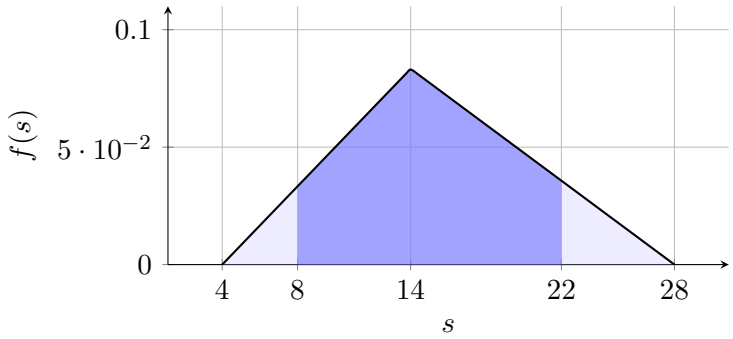
The interval straddles the mode; both tails must be removed:

$$P(8 \le S \le 22) = 1 - \frac{1}{2}(8-4)f(8) - \frac{1}{2}(28-22)f(22).$$

Here $f(8) = (8-4)/120 = \frac{1}{30}$ and $f(22) = (28-22)/168 = \frac{1}{28}$ min^{-1}.

$$\begin{aligned} P &= 1 - \tfrac{1}{2}(4)(\frac{1}{30}) - \tfrac{1}{2}(6)(\frac{1}{28}) \\ &= 1 - \frac{1}{15} - \frac{3}{28} = \boxed{\frac{347}{420} \approx 0.8262 = 82.62\%}. \end{aligned}$$

Shaded event: $8 \le S \le 22$; s in minutes



Density has units min^{-1}. Integrating over $[8, 14]$ and $[14, 22]$ gives the same probability.

Return to this task

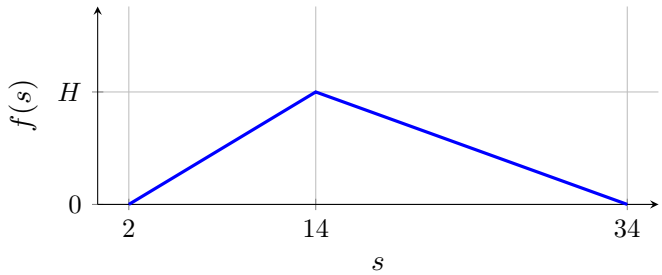
Solution: Changed range

What stays fixed. The mode is 14 min, the event is $[8, 22]$, and both branches remain straight. Only the support endpoints change.

Stage 1: Support and shape

$S \sim \text{Tri}(2, 34, 14)$ has support $[2, 34]$ min. The density rises from $(2, 0)$ to $(14, H)$ and falls to $(34, 0)$.

Support and triangular shape; horizontal coordinate in minutes



Stage 2: Total area = 1

The triangle area is $\frac{1}{2} \times$ base $\times$ height, so

$$1 = \frac{1}{2}(34-2)H \quad \Longrightarrow \quad H = \frac{2}{32} = \frac{1}{16}\ \text{min}^{-1}.$$

Stage 3: Complete piecewise pdf

Use $m = (y_2 - y_1)/(x_2 - x_1)$ and $q = y_1 - mx_1$ in $f(s) = ms + q$. For the rising line through $(2, 0)$ and $(14, \frac{1}{16})$,

$$m_1 = \frac{\frac{1}{16} - 0}{14 - 2} = \frac{1}{192}, \qquad q_1 = 0 - (\frac{1}{192})(2) = \frac{-1}{96}, \qquad f_1(s) = \frac{s-2}{192}.$$

For the falling line through $(14, \frac{1}{16})$ and $(34, 0)$,

$$m_2 = \frac{0 - \frac{1}{16}}{34 - 14} = \frac{-1}{320}, \qquad q_2 = 0 - (\frac{-1}{320})(34) = \frac{17}{160}.$$

Thus the complete density is

$$f(s) = \begin{cases} \dfrac{s-2}{192}, & 2 \le s \le 14, \\ \dfrac{34-s}{320}, & 14 < s \le 34, \\ 0, & \text{otherwise.} \end{cases}$$

The branches meet at height $\frac{1}{16}$, are nonnegative, and enclose total area 1.

Stage 4: Probability by complementary tail triangles

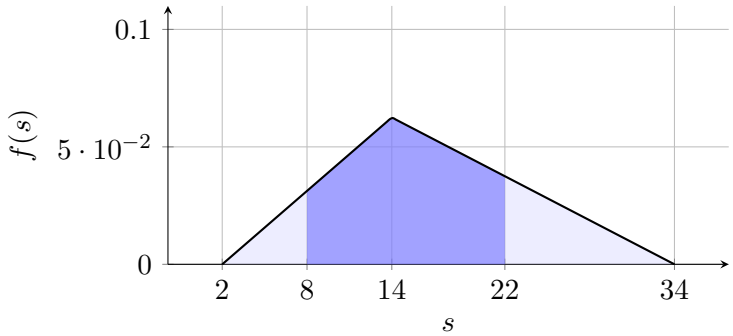
The interval straddles the mode; both tails must be removed:

$$P(8 \le S \le 22) = 1 - \frac{1}{2}(8-2)f(8) - \frac{1}{2}(34-22)f(22).$$

Here $f(8) = (8-2)/192 = \frac{1}{32}$ and $f(22) = (34-22)/320 = \frac{3}{80}\ \text{min}^{-1}$.

$$\begin{aligned} P &= 1 - \tfrac{1}{2}(6)(\frac{1}{32}) - \tfrac{1}{2}(12)(\frac{3}{80}) \\ &= 1 - \frac{3}{32} - \frac{9}{40} = \boxed{\frac{109}{160} \approx 0.6813 = 68.13\%}. \end{aligned}$$

Shaded event: $8 \le S \le 22$**;** s **in minutes**



Density has units min^{-1}. Integrating over $[8, 14]$ and $[14, 22]$ gives the same probability.

Return to this task

3 A repeating irrigation cycle [C3-3]

A pressure fluctuation at a Colombian flower farm occurs at a uniformly distributed position within a repeating 48-minute irrigation cycle. The cycle starts at minute 0 and restarts immediately after minute 48.

Questions/Tasks

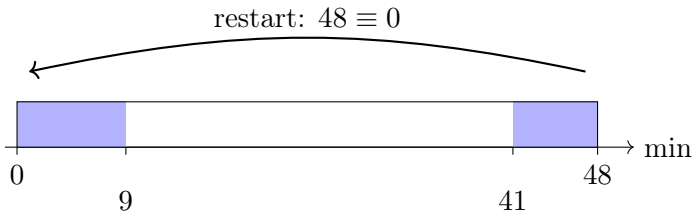
C3 Find the probability that the fluctuation occurs in the window from minute 41 through the restart to minute 9 of the next cycle. **[1]**

C3 Find the probability that the fluctuation occurs within 150 seconds before or after the start of a cycle. **[2]**

Solution: Window through the restart

Let $U \sim U(0, 48)$ represent position in minutes, with density $1/48$ min^{-1} on $[0, 48)$. The window splits into $[41, 48)$ and $[0, 9]$, of lengths 7 and 9 minutes.

The shaded lengths meet through the restart



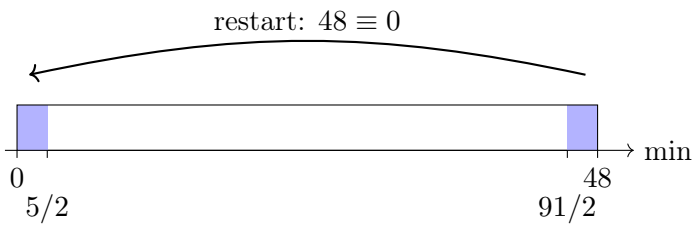
$$P = \frac{(48 - 41) + (9 - 0)}{48} = \frac{1}{3} \approx 0.3333 = 33.33\%\,.$$

Return to this task

Solution: Tolerance on both sides

Convert first: 150 s $= 150/60 = 5/2$ min on *each* side. The intervals are $[0, 5/2]$ and $[48 - 5/2, 48) = [91/2, 48)$.

Two equal pieces beside the cycle start



$$P = \frac{5/2 + 5/2}{48} = \frac{5}{48} \approx 0.1042 = 10.42\%\,.$$

Endpoints have probability zero; their inclusion does not change either answer.

Return to this task

4 Direction of a floating current marker [C3-3]

A river-monitoring team in Colombia releases a floating marker. Its initial direction is uniformly distributed around a full circle. A bank marker points north. "Within an angle" means that tolerance on **either side** of north, including wraparound.

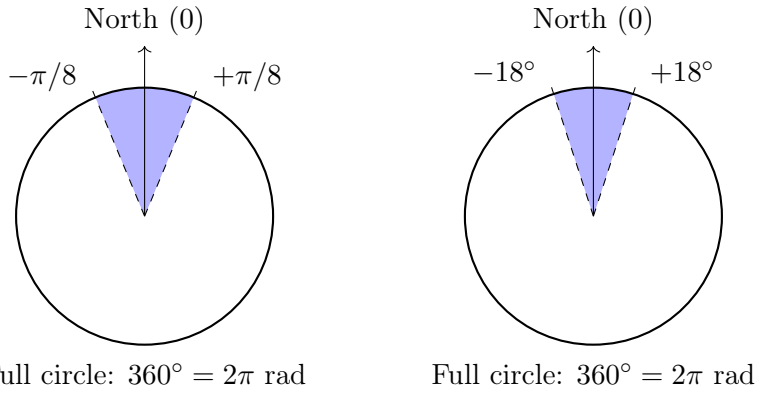
Questions/Tasks

C3 Compare the probability of a direction within $\pi/8$ radians of north with that of a direction within $18°$ of north. Calculate both probabilities and state which exceeds the other and by how much. **[3]**

Solution: Angular probabilities and comparison

Uniform directions give probability equal to favourable angle divided by full-circle angle. Count *both sides* of north.

Two wraparound sectors centred at north



In radians, the event is $[0, \pi/8] \cup [2\pi - \pi/8, 2\pi)$:

$$P_{\pi/8} = \frac{2(\pi/8)}{2\pi} = \boxed{\frac{1}{8} = 0.1250 = 12.50\%}\,.$$

In degrees, the event is $[0, 18] \cup [342, 360)$:

$$P_{18°} = \frac{2(18°)}{360°} = \boxed{\frac{1}{10} = 0.1000 = 10.00\%}\,.$$

The $\pi/8$ tolerance (22.5 degrees on each side) gives greater probability:

$$\Delta P = \frac{1}{8} - \frac{1}{10} = \boxed{\frac{1}{40} = 0.0250}\,,$$

an absolute difference of **2.50 percentage points**. Using the 18-degree probability as baseline, the relative increase is

$$100\frac{1/40}{1/10}\% = 25.00\%.$$

Return to this task

5 Public-library return arrivals [C4-3]

Returns enter an automated public-library return chute according to a Poisson process. The rate differs between the four scenarios below but is constant within each scenario. From a fixed observation time, let T be the waiting time in minutes for the next return.

Questions/Tasks

C4 With an average of 7 returns every 35 minutes, find $P(T < 3)$, the probability of waiting less than 3 minutes. **[0.5]**

C4 With a mean waiting time of 5 minutes between returns, find $P(T > 8)$. **[0.5]**

C4 With an average of 4 returns every 25 minutes, find $P(1 < T < 6)$. **[1]**

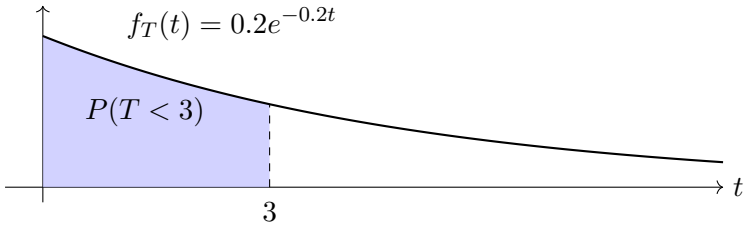
C4 With an average of 18 returns per hour, find $P(T < 2 \text{ or } T > 7)$. **[1]**

Solution: Less than three minutes

The rate is $\lambda_1 = 7/35 = 1/5 = 0.200$ returns/min. For $T \sim \text{Exp}(\lambda_1)$, $P(T < t) = 1 - e^{-\lambda_1 t}$.

$$P(T < 3) = 1 - e^{-(1/5)(3)} = 1 - e^{-3/5} \approx 0.4512 = 45.12\%.$$

Shaded event; t in minutes and $f_T(t)$ in inverse minutes



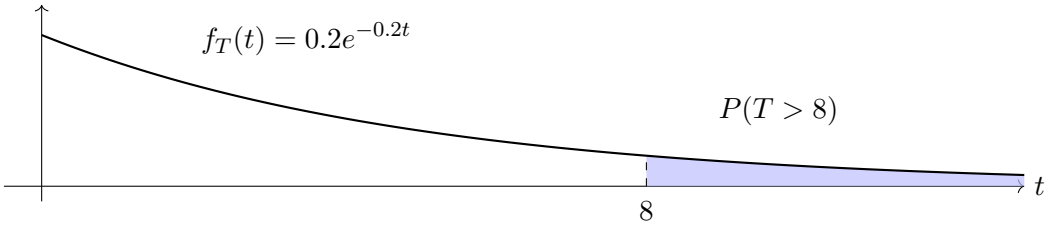
Return to this task

Solution: More than eight minutes

A mean wait of 5 min gives $\lambda_2 = 1/E(T) = 1/5 = 0.200$ returns/min. Use the survival formula $P(T > t) = e^{-\lambda_2 t}$.

$$P(T > 8) = e^{-(1/5)(8)} = \boxed{e^{-8/5} \approx 0.2019 = 20.19\%}.$$

Shaded event; t in minutes and $f_T(t)$ in inverse minutes



The shaded right tail continues beyond the plotted edge to infinity.

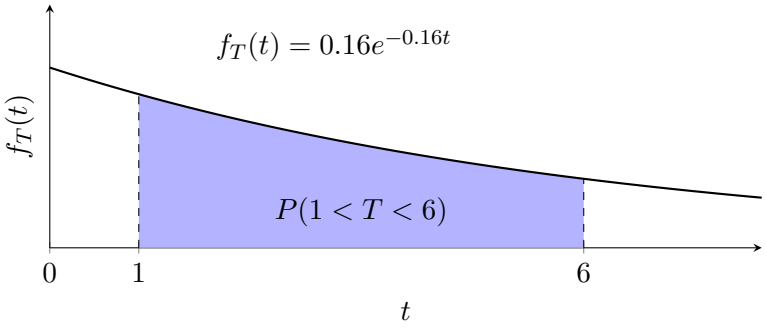
Return to this task

Solution: Between one and six minutes

The separate rate is $\lambda_3 = 4/25 = 0.160$ returns/min. For an interval, $P(a < T < b) = e^{-\lambda_3 a} - e^{-\lambda_3 b}$.

$$P(1 < T < 6) = e^{-(4/25)(1)} - e^{-(4/25)(6)} = \boxed{e^{-4/25} - e^{-24/25} \approx 0.4693 = 46.93\%}.$$

Shaded event; t in minutes and $f_T(t)$ in inverse minutes

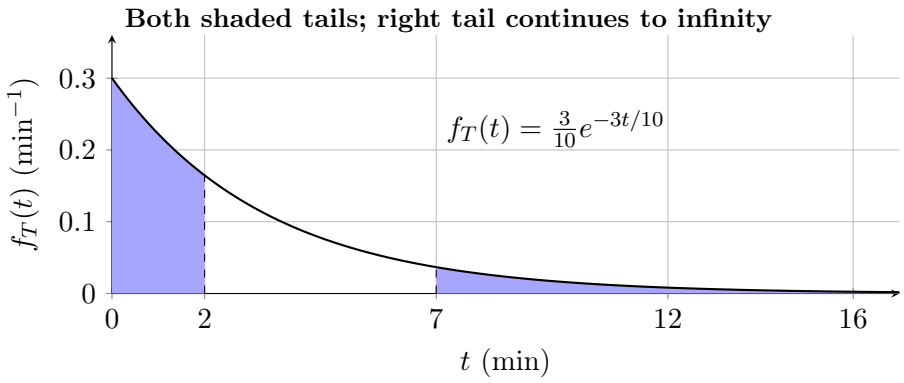


Return to this task

Solution: Outside two to seven minutes

Convert the hourly rate: $\lambda_4 = 18/60 = 3/10 = 0.300$ returns/min. The two tails are disjoint, so $P(T < a \text{ or } T > b) = (1 - e^{-\lambda_4 a}) + e^{-\lambda_4 b}$.

$$P(T < 2 \text{ or } T > 7) = 1 - e^{-(3/10)(2)} + e^{-(3/10)(7)} = \boxed{1 - e^{-3/5} + e^{-21/10} \approx 0.5736 = 57.36\%}.$$



For all four scenarios, $f_T(t) = \lambda e^{-\lambda t}$ for $t \geq 0$, zero otherwise, with support $[0, \infty)$. The constant-rate Poisson assumption makes exponential waiting times appropriate.

Return to this task

6 Bicycle-hire arrival rates [C4-2]

At a municipal bicycle-hire kiosk, rental starts follow a Poisson process with a constant rate during each shift. Morning and afternoon rates differ. From a fixed time within each shift, let T_m, T_a be waiting times in minutes for the next rental start. Assume each relevant shift continues throughout the waiting time.

Questions/Tasks

C4 In the morning, $P(T_m > 12) = 0.35$. Infer the rate in rental starts per hour. **[1]**

C4 In the afternoon, $P(T_a < 9) = 0.72$. Infer the rate in rental starts per hour. **[1]**

Solution: Morning rate from survival

Use $P(T_m > t) = e^{-\lambda_m t}$:

$$0.35 = e^{-12\lambda_m} \implies \ln(0.35) = -12\lambda_m \implies \lambda_m = -\frac{\ln(0.35)}{12} \approx 0.087 \text{ starts/min}.$$

Convert the exact rate to hourly units:

$$60\lambda_m = -5\ln(0.35) \approx \boxed{5.249 \text{ starts/hour}}.$$

Return to this task

Solution: Afternoon rate from a cumulative probability

Use $P(T_a < t) = 1 - e^{-\lambda_a t}$:

$$0.72 = 1 - e^{-9\lambda_a} \implies e^{-9\lambda_a} = 0.28 \implies -9\lambda_a = \ln(0.28).$$

$$\lambda_a = -\frac{\ln(0.28)}{9} \approx 0.141 \text{ starts/min}, \qquad 60\lambda_a = -\frac{20}{3}\ln(0.28) \approx \boxed{8.486 \text{ starts/hour}}.$$

Return to this task

7 Two recycling collection gates [C4-2]

Vehicle entries at gates A and B of a Colombian recycling facility follow independent constant-rate Poisson processes. From a fixed observation time, the probability of A's next entry within 4 minutes is 0.84; the probability that B's next entry takes more than 4 minutes is 0.60. Count entries at both gates from this same observation time.

Questions/Tasks

C4 Infer both rates. After how many minutes does A's expected entry count exceed B's expected count by 4? Find the time for this difference in expectations. **[2]**

Solution: Rates and expected-count difference

For A the given probability is cumulative:

$$0.84 = 1 - e^{-4\lambda_A} \implies e^{-4\lambda_A} = 0.16 \implies \lambda_A = -\frac{\ln(0.16)}{4} \approx 0.458 \text{ entries/min}.$$

For B it is a survival probability:

$$0.60 = e^{-4\lambda_B} \implies \lambda_B = -\frac{\ln(0.60)}{4} \approx 0.128 \text{ entries/min}.$$

Define $N_A(t), N_B(t)$ as entry counts since the common observation time. For constant-rate Poisson processes,

$$E[N_A(t)] = \lambda_A t, \qquad E[N_B(t)] = \lambda_B t.$$

Thus $E[N_A(t)] - E[N_B(t)] = (\lambda_A - \lambda_B)t$. Since $\lambda_A > \lambda_B$, the required time is positive and finite:

$$t = \frac{4}{\lambda_A - \lambda_B} = \frac{16}{\ln(0.60/0.16)} = \boxed{\frac{16}{\ln(15/4)} \approx 12.105 \text{ min}}.$$

This specifies a difference of 4 in *expected* counts; the actual random count difference need not be 4 at this time.

Return to this task

8 Two document-digitisation desks [C5-2]

At a Colombian municipal archive, let A and B be the times in minutes to digitise a file at two desks. Both have support $[2, 17]$. At A the density increases along one straight line from zero at 2 to its highest value at 17. At B, equal-length intervals within the support have equal probabilities. Both densities are zero outside the support.

Questions/Tasks

C5 Identify A's distribution and calculate its mean, variance and standard deviation. **[1]**

C5 Identify B's distribution and calculate its mean, variance and standard deviation. **[1]**

Solution: Increasing-linear digitisation time

The increasing straight density is an endpoint triangular model: $A \sim \text{Tri}(2, 17, 17)$, support $[2, 17]$ min. For increasing-linear density on $[a, b]$,

$$\mu = \frac{a + 2b}{3}, \qquad \sigma^2 = \frac{(b-a)^2}{18}, \qquad \sigma = \sqrt{\sigma^2}.$$

$$E(A) = \frac{2 + 2(17)}{3} = 12 \text{ min}, \qquad \text{Var}(A) = \frac{(17-2)^2}{18} = \frac{25}{2} = 12.500 \text{ min}^2,$$

$$\sigma_A = \sqrt{25/2} \approx 3.536 \text{ min}.$$

Over many files, desk A averages 12 minutes per file.

Return to this task

Solution: Uniform digitisation time

Equal-length intervals have equal probabilities, so $B \sim U(2, 17)$, support $[2, 17]$ min. The formulas are $\mu = (a + b)/2$, $\sigma^2 = (b - a)^2/12$, $\sigma = \sqrt{\sigma^2}$.

$$E(B) = \frac{2 + 17}{2} = 9.500 \text{ min}, \qquad \text{Var}(B) = \frac{(17-2)^2}{12} = \frac{75}{4} = 18.750 \text{ min}^2,$$

$$\sigma_B = \sqrt{75/4} \approx 4.330 \text{ min}.$$

Desk B averages 9.5 minutes per file; its time SD is about 4.330 minutes.

Return to this task

9 Pump maintenance and daily operating costs [C5-2]

A rural water association models pump maintenance duration D on $[2, 14]$ hours. Its density decreases along one straight line from its highest value at 2 to zero at 14 and is zero outside the support.

Daily operating cost R, in **thousands of Colombian pesos (COP)**, has an exact symmetric bell-shaped density on the real line, centred at 2400, with inflection points 2250 and 2550, all in thousands of COP.

Questions/Tasks

C5 Identify D's distribution and calculate its mean, variance and standard deviation. **[1]**

C5 Identify R's distribution and infer its parameters. Calculate mean and SD in **COP** and variance in **COP**^{2}. Interpret the mean. **[1]**

Solution: Decreasing-linear maintenance duration

The decreasing straight density gives $D \sim \text{Tri}(2, 14, 2)$, support $[2, 14]$ h. Use $\mu = (2a + b)/3$, $\sigma^2 = (b - a)^2/18$, $\sigma = \sqrt{\sigma^2}$:

$$E(D) = \frac{2(2) + 14}{3} = 6 \text{ h}, \qquad \text{Var}(D) = \frac{(14 - 2)^2}{18} = 8 \text{ h}^2, \qquad \sigma_D = \sqrt{8} \approx 2.828 \text{ h}.$$

The model predicts a long-run average maintenance duration of 6 hours.

Return to this task

Solution: Normal cost and conversion to COP

The exact symmetric bell curve identifies a normal distribution. In thousands of COP, the centre is $\mu_R = 2400$. Normal inflection points are $\mu_R \pm \sigma_R$, so

$$\sigma_R = 2550 - 2400 = 2400 - 2250 = 150, \qquad R \sim N(2400, 150^2) = N(2400, 22500).$$

Its mathematical support is $\mathbb{R}$. In these units, mean is μ_R, variance is σ_R^2, and SD is σ_R. Let $Y = 1000R$ be cost in COP. The conversion laws are

$$E(cR) = cE(R), \quad \text{Var}(cR) = c^2 \,\text{Var}(R), \quad \text{SD}(cR) = |c| \,\text{SD}(R).$$

Therefore

$$E(Y) = 1000(2400) = 2{,}400{,}000 \text{ COP},$$

$$\text{Var}(Y) = 1000^2(22500) = 22{,}500{,}000{,}000 \text{ COP}^2, \qquad \text{SD}(Y) = 1000(150) = 150{,}000 \text{ COP}.$$

The association's long-run average daily operating cost is 2,400,000 COP; individual days vary around it.

Return to this task

10 Museum admissions and audio-guide sessions [C5-2]

Admissions to a Colombian museum follow a constant-rate Poisson process averaging 15 admissions per hour. Let T be interarrival time in minutes.

Audio-guide session duration S has support $[6, 27]$ minutes. Its density rises linearly from zero at 6 to a single peak at 15, then falls linearly to zero at 27 and is zero outside this range.

Questions/Tasks

C5 Identify T's distribution and calculate its mean and SD in minutes and its variance in minutes^2. **[1]**

C5 Identify S's distribution and calculate its mean and SD in minutes and its variance in minutes^2. **[1]**

Solution: Exponential interarrival time

The constant-rate Poisson process gives $T \sim \text{Exp}(\lambda)$, support $[0, \infty)$ min, with $\lambda = 15/60 = 1/4 = 0.250$ admissions/min. Use $\mu = 1/\lambda$, $\sigma^2 = 1/\lambda^2$, $\sigma = 1/\lambda$:

$$E(T) = \frac{1}{1/4} = 4 \text{ min}, \qquad \text{Var}(T) = \frac{1}{(1/4)^2} = 16 \text{ min}^2, \qquad \sigma_T = 4 \text{ min}.$$

The mean gap between admissions is 4 minutes.

Return to this task

Solution: Triangular audio-guide duration

The straight branches give $S \sim \text{Tri}(6, 27, 15)$, support $[6, 27]$ min. For minimum a, maximum b, mode c,

$$\mu = \frac{a+b+c}{3}, \quad \sigma^2 = \frac{a^2+b^2+c^2-ab-ac-bc}{18}, \quad \sigma = \sqrt{\sigma^2}.$$

$$E(S) = \frac{6+27+15}{3} = 16 \text{ min}.$$

$$\text{Var}(S) = \frac{6^2+27^2+15^2-6(27)-6(15)-27(15)}{18}$$

$$= \frac{333}{18} = \frac{37}{2} = 18.500 \text{ min}^2,$$

$$\sigma_S = \sqrt{37/2} \approx 4.301 \text{ min}.$$

The model predicts a mean session of 16 minutes with SD about 4.301 minutes.

Return to this task

11 Commercial-kitchen water use [C6-2]

A Colombian catering kitchen records water use X in litres for 50 cleaning batches, in two non-overlapping parts. The first 20 observations are:

4.1	4.5	5.0	5.3	5.8
6.0	6.2	6.7	7.0	7.4
4.8	5.0	5.5	5.9	6.4
6.8	7.0	7.2	7.6	8.0

The remaining 30 observations are summarised in the discrete frequency table:

x_i	5.0	5.4	5.8	6.3	7.0	7.5
f_i	4	6	5	7	3	5

The model gives $P(5 \le X < 7) = 0.58$; include 5 and exclude 7. Combine both parts as one sample.

Questions/Tasks

C6 Find the empirical probability of this event. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

Eligible raw observations are $5.0, 5.3, 5.8, 6.0, 6.2, 6.7, 5.0, 5.5, 5.9, 6.4, 6.8$, giving **11** batches. Table values $5.0, 5.4, 5.8, 6.3$ contribute $4 + 6 + 5 + 7 = 22$ batches. Include every 5 and exclude every 7. The records do not overlap, so $n = 20 + 30 = 50$ and $k = 11 + 22 = 33$.

$$\widehat{P}(5 \le X < 7) = \frac{33}{50} = 0.6600 = 66.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$, using the exact relative frequency:

$$E = \left|\frac{33}{50} - 0.58\right| = \frac{2}{25} \approx 0.0800.$$

$$E\% = 100\frac{\frac{2}{25}}{0.58}\% \approx 13.79\%.$$

Probabilities and their errors are unitless.

Return to this task

12 Produce cooling, grouped data [C6-2]

Cooling times X in minutes for produce crates in a Colombian packing centre are grouped below. Classes include the left endpoint and exclude the right endpoint.

Class	$[2,7)$	$[7,12)$	$[12,17)$	$[17,22)$
f_i	4	7	6	4

The model gives $P(7 \le X < 17) = 0.68$.

Questions/Tasks

C6 Find the empirical probability of this event. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

The complete included classes are $[7,12)$ and $[12,17)$. Thus $n = 4+7+6+4 = 21$ and $k = 7+6 = 13$. These counts are exact because event endpoints are class boundaries; no midpoint approximation is needed.

$$\widehat{P}(7 \le X < 17) = \frac{13}{21} \approx 0.6190 = 61.90\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$, using the exact relative frequency:

$$E = \left|\frac{13}{21} - 0.68\right| = \frac{32}{525} \approx 0.0610.$$

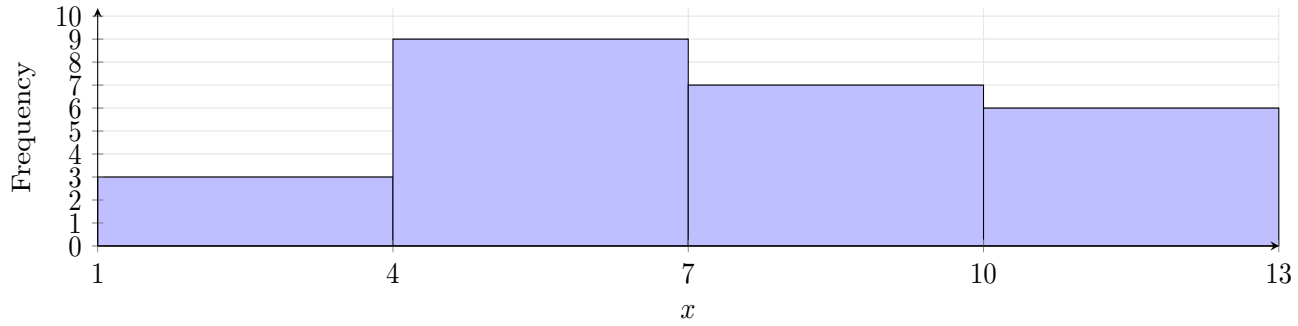
$$E\% = 100\frac{\frac{32}{525}}{0.68}\% \approx 8.96\%.$$

Probabilities and their errors are unitless.

Return to this task

13 Battery-swap service, histogram [C6-2]

The histogram is the only supplied sample record of battery-swap service times X in minutes at a Colombian electric-motorcycle station. Equal-width classes include the left endpoint and exclude the right endpoint. Bar heights are frequencies; the horizontal axis gives minutes.



The model gives $P(4 \le X < 10) = 0.60$.

Questions/Tasks

C6 Find the empirical probability of this event. **[1]**

C6 Calculate its absolute and percentage errors. **[1]**

Solution: Empirical probability

Read the four histogram heights as frequencies $3, 9, 7, 6$. The included complete classes are $[4, 7)$ and $[7, 10)$, giving $n = 3 + 9 + 7 + 6 = 25$ and $k = 9 + 7 = 16$. No midpoint approximation is needed.

$$\widehat{P}(4 \le X < 10) = \frac{16}{25} = 0.6400 = 64.00\%.$$

Return to this task

Solution: Absolute and percentage errors

Use $E = |\widehat{P} - P_0|$ and $E\% = 100E/|P_0|$, using the exact relative frequency:

$$E = \left|\frac{16}{25} - 0.60\right| = \frac{1}{25} \approx 0.0400.$$

$$E\% = 100\frac{\frac{1}{25}}{0.60}\% \approx 6.67\%.$$

Probabilities and their errors are unitless.

Return to this task

14 Daily paper use, mixed data [C7-2]

A Colombian legal-aid office records paper use in hundreds of sheets on 13 days, in two non-overlapping parts. The first five observations are $13, 16, 18, 21, 23$. The remaining eight are summarised below:

x_i	13	18	23
f_i	1	4	3

Combine both parts as one sample. The theoretical mean is 18 hundreds of sheets and variance is 16 (hundreds of sheets)^2.

Questions/Tasks

C7 Calculate mean, variance and standard deviation. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from combined data

The five raw observations contribute one each at $13, 16, 18, 21, 23$; the table adds $1, 4, 3$ at $13, 18, 23$. The combined actual-value frequencies below require no midpoint estimation.

x_i	13	16	18	21	23
f_i	2	1	5	1	4

Use sample variance with denominator $n-1$:

$$n = \sum f_i = 13, \qquad \bar{x} = \frac{\sum f_i x_i}{n}.$$

$$\bar{x} = \frac{2(13) + 1(16) + 5(18) + 1(21) + 4(23)}{13} = \frac{245}{13} \approx 18.846 \text{ hundreds of sheets.}$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (x_i - \bar{x})^2 \\ &= 2\left(13 - \frac{245}{13}\right)^2 + 1\left(16 - \frac{245}{13}\right)^2 + 5\left(18 - \frac{245}{13}\right)^2 \\ &\quad + 1\left(21 - \frac{245}{13}\right)^2 + 4\left(23 - \frac{245}{13}\right)^2 \\ &= \frac{1998}{13} \text{ (hundreds of sheets)}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{n-1} = \frac{\frac{1998}{13}}{13-1} = \frac{333}{26} \approx 12.808 \text{ (hundreds of sheets)}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{333}{26}} \approx 3.579 \text{ hundreds of sheets.}$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$, $E\% = 100E/|z_0|$. Using unrounded statistics:

$$E_{\bar{x}} = \left|\frac{245}{13} - 18\right| = \frac{11}{13} \approx 0.846 \text{ hundreds of sheets}, \qquad E_{\bar{x}}\% = 100\frac{\frac{11}{13}}{18}\% \approx 4.70\%.$$

$$E_{s^2} = \left|\frac{333}{26} - 16\right| = \frac{83}{26} \approx 3.192 \text{ (hundreds of sheets)}^2, \qquad E_{s^2}\% = 100\frac{\frac{83}{26}}{16}\% \approx 19.95\%.$$

The requested errors concern the mean and variance only.

Return to this task

15 Compost maturation checks, grouped data [C7-2]

A Colombian urban garden groups the times in days between compost checks. Classes include the left endpoint and exclude the right endpoint. The theoretical mean is 15 days and variance is 24 days^2.

Class	$[5, 11)$	$[11, 17)$	$[17, 23)$
f_i	3	2	2

Questions/Tasks

C7 Calculate mean, variance and standard deviation. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from class midpoints

The class midpoint is (left + right)/2. Individual observations are unavailable, so all three statistics and their errors are **midpoint estimates**.

m_i	8	14	20
f_i	3	2	2

Use sample variance with denominator $n-1$:

$$n = \sum f_i = 7, \qquad \bar{x} = \frac{\sum f_i m_i}{n}.$$

$$\bar{x} \approx \frac{3(8) + 2(14) + 2(20)}{7} = \frac{92}{7} \approx 13.143 \text{ days}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (m_i - \bar{x})^2 \\ &= 3\left(8 - \frac{92}{7}\right)^2 + 2\left(14 - \frac{92}{7}\right)^2 + 2\left(20 - \frac{92}{7}\right)^2 \\ &= \frac{1224}{7} \text{ (days)}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1} \approx \frac{\frac{1224}{7}}{7-1} = \frac{204}{7} \approx 29.143 \text{ (days)}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{204}{7}} \approx 5.398 \text{ days}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$, $E\% = 100E/|z_0|$. Using unrounded midpoint estimates:

$$E_{\bar{x}} = \left|\frac{92}{7} - 15\right| = \frac{13}{7} \approx 1.857 \text{ days}, \qquad E_{\bar{x}}\% = 100\frac{\frac{13}{7}}{15}\% \approx 12.38\%.$$

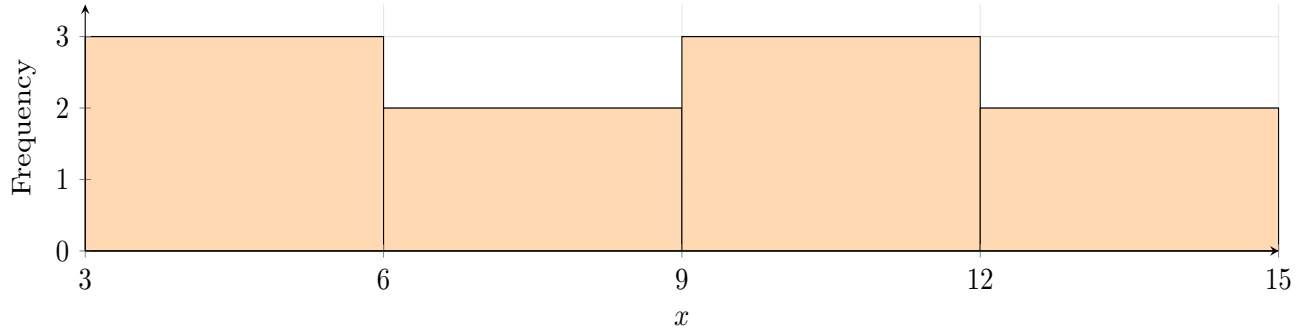
$$E_{s^2} = \left|\frac{204}{7} - 24\right| = \frac{36}{7} \approx 5.143 \text{ (days)}^2, \qquad E_{s^2}\% = 100\frac{\frac{36}{7}}{24}\% \approx 21.43\%.$$

The requested errors concern the mean and variance only.

Return to this task

16 Community-bakery flour use, histogram [C7-2]

The histogram is the only supplied sample record of flour use per production batch in kilograms at a Colombian community bakery. Equal-width classes include the left endpoint and exclude the right endpoint. Bar heights are frequencies; the horizontal axis gives kilograms. The theoretical mean is 8 kg and variance is 7 kg^2.



Questions/Tasks

C7 Calculate mean, variance and standard deviation. **[1]**

C7 Calculate absolute and percentage errors for mean and variance. **[1]**

Solution: Sample statistics from class midpoints

The class midpoint is (left + right)/2. Individual observations are unavailable, so all three statistics and their errors are **midpoint estimates**.

m_i	$\frac{9}{2}$	$\frac{15}{2}$	$\frac{21}{2}$	$\frac{27}{2}$
f_i	3	2	3	2

Use sample variance with denominator $n-1$:

$$n = \sum f_i = 10, \qquad \bar{x} = \frac{\sum f_i m_i}{n}.$$

$$\bar{x} \approx \frac{3(\frac{9}{2}) + 2(\frac{15}{2}) + 3(\frac{21}{2}) + 2(\frac{27}{2})}{10} = \frac{87}{10} \approx 8.700 \text{ kg}.$$

The weighted squared-deviation sum is

$$\begin{aligned} Q &= \sum_i f_i (m_i - \bar{x})^2 \\ &= 3\left(\frac{9}{2} - \frac{87}{10}\right)^2 + 2\left(\frac{15}{2} - \frac{87}{10}\right)^2 + 3\left(\frac{21}{2} - \frac{87}{10}\right)^2 \\ &\quad + 2\left(\frac{27}{2} - \frac{87}{10}\right)^2 \\ &= \frac{558}{5} \text{ (kg)}^2. \end{aligned}$$

$$s^2 = \frac{\sum_i f_i (m_i - \bar{x})^2}{n-1} \approx \frac{\frac{558}{5}}{10-1} = \frac{62}{5} \approx 12.400 \text{ (kg)}^2,$$

$$s = \sqrt{s^2} \approx \sqrt{\frac{62}{5}} \approx 3.521 \text{ kg}.$$

Return to this task

Solution: Mean and variance errors

For a statistic z and theoretical value z_0, use $E = |z - z_0|$, $E\% = 100E/|z_0|$. Using unrounded midpoint estimates:

$$E_{\bar{x}} = \left|\frac{87}{10} - 8\right| = \frac{7}{10} \approx 0.700 \text{ kg}, \qquad E_{\bar{x}}\% = 100\frac{\frac{7}{10}}{8}\% \approx 8.75\%.$$

$$E_{s^2} = \left|\frac{62}{5} - 7\right| = \frac{27}{5} \approx 5.400 \text{ (kg)}^2, \qquad E_{s^2}\% = 100\frac{\frac{27}{5}}{7}\% \approx 77.14\%.$$

The requested errors concern the mean and variance only.

Return to this task

17 Summarized Solutions

Exact parameters and counts are retained. Linked problems provide procedures and supports.

C1 – All twelve variables

Problem 1. D = discrete; C = continuous.

Context	Variable	Class	Model (parameters as defined above)
1	a	D	Bernoulli(0.86)
1	b	D	Binomial$(35, 0.86)$
2	a	D	Geometric(0.06), including the first faulty meter
2	b	C	$U(-2, 4)$ L
3	a	D	Poisson(3) bookings in one hour
3	b	C	Tri$(8, 26, 15)$ min
4	a	C	Increasing-linear: $200(x - 0.2)/49$ on $[0.2, 0.9]$
4	b	C	$N(1, 9)$ in watts
5	a	D	Binomial$(45, 0.07)$
5	b	D	Poisson(24) calls over four days
6	a	C	Candidate Tri$(12, 36, 12)$ min if linearity is assumed
6	b	C	Candidate $N(18, 6.25)$ in kg if normality is assumed

Context 6 is not uniquely determined: steady decrease need not be linear, and symmetry with SD 2.5 kg need not be normal. The normal candidate has support $\mathbb{R}$; physical use is nonnegative, with exact support unspecified.

C2 – Triangular models

Problem 2. Mode 14 min and event $[8, 22]$ stay fixed. Densities are zero outside their supports.

Support	Rising pdf	Falling pdf	$P(8 \le S \le 22)$
$[4, 28]$	$(s - 4)/120$	$(28 - s)/168$	$\frac{347}{420} \approx 0.8262 = 82.62\%$
$[2, 34]$	$(s - 2)/192$	$(34 - s)/320$	$\frac{109}{160} \approx 0.6813 = 68.13\%$

Rising branches end at 14; falling branches start after 14. Heights are $1/12$ and $1/16$ min^{-1}.

C3 – Cycles and directions

Problem 3: $[41, 48) \cup [0, 9]$ gives $1/3 \approx 0.3333 = 33.33\%$. The 150-second tolerance on each side gives $[0, 5/2] \cup [91/2, 48)$ and $5/48 \approx 0.1042 = 10.42\%$.

Problem 4: $P_{\pi/8} = 1/8 = 0.1250 = 12.50\%$; $P_{18^\circ} = 1/10 = 0.1000 = 10.00\%$. The first exceeds the second by $1/40 = 0.0250$, or 2.50 percentage points (relative increase 25.00%).

C4 – Waiting probabilities and inferred rates

Problem 5. Rates are returns/minute.

Scenario	Rate	Exact probability	Decimal and percentage
1	$\frac{1}{5}$	$1 - e^{-3/5}$	0.4512; 45.12%
2	$\frac{1}{5}$	$e^{-8/5}$	0.2019; 20.19%
3	$\frac{4}{25}$	$e^{-4/25} - e^{-24/25}$	0.4693; 46.93%
4	$\frac{3}{10}$	$1 - e^{-3/5} + e^{-21/10}$	0.5736; 57.36%

Events in order: $T < 3$, $T > 8$, $1 < T < 6$, $T < 2$ or $T > 7$. Problem 6: morning $\lambda_m = -\ln(0.35)/12 \approx 0.087$ starts/min, $60\lambda_m \approx 5.249$ starts/hour. Afternoon $\lambda_a = -\ln(0.28)/9 \approx 0.141$ starts/min, $60\lambda_a \approx 8.486$ starts/hour.

Problem 7: $\lambda_A = -\ln(0.16)/4 \approx 0.458$ and $\lambda_B = -\ln(0.60)/4 \approx 0.128$ entries/min. The expected-count difference is 4 at $t = 16/\ln(15/4) \approx 12.105$ min; actual counts remain random.

C5 – Theoretical moments

Supports: $A, B \in [2, 17]$ min; $D \in [2, 14]$ h; normal cost has support $\mathbb{R}$; $T \in [0, \infty)$ min; session $S \in [6, 27]$ min.

Problem	Variable/model	Mean	Variance	SD	Units
8	$A \sim \text{Tri}(2, 17, 17)$	12	12.500	3.536	min; min^2
8	$B \sim U(2, 17)$	9.500	18.750	4.330	min; min^2
9	$D \sim \text{Tri}(2, 14, 2)$	6	8	2.828	h; h^2
10	$T \sim \text{Exp}(1/4)$	4	16	4	min; min^2
10	$S \sim \text{Tri}(6, 27, 15)$	16	18.500	4.301	min; min^2

Units are first for mean and SD, second for variance. Exact session moments: 16, $37/2$, $\sqrt{37/2}$. Problem 9: $R \sim N(2400, 22500)$ in thousands of COP, SD 150. In COP: mean 2,400,000, variance 22,500,000,000 COP^2, SD 150,000 COP. The mean is the long-run average daily operating cost.

C6 – Empirical probabilities

$E = |\widehat{P} - P_0|$, $E\% = 100E/|P_0|$. Complete event classes give exact counts.

Problem	Event	n	k	$\widehat{P}$	P_0	E	$E\%$
11	$5 \le X < 7$ L	50	33	0.6600	0.5800	0.0800	13.79%
12	$7 \le X < 17$ min	21	13	0.6190	0.6800	0.0610	8.96%
13	$4 \le X < 10$ min	25	16	0.6400	0.6000	0.0400	6.67%

Water-use event count: 11 raw plus 22 table observations, including 5 and excluding 7.

C7 – Sample statistics and requested errors

All sample variances use $n-1$. Asterisks mark midpoint estimates because individual observations are unavailable.

Problem	n	$\bar{x}$	s^2	s	Units
14	13	18.846	12.808	3.579	hundreds of sheets; squared
15*	7	13.143	29.143	5.398	days; $(\text{days})^2$
16*	10	8.700	12.400	3.521	kg; $(\text{kg})^2$

Exact statistics $(\bar{x}, s^2)$, in problem order: $(\frac{245}{13}, \frac{333}{26})$, $(\frac{92}{7}, \frac{204}{7})$, $(\frac{87}{10}, \frac{62}{5})$.

Problem	Statistic	Absolute error	Percentage error	Absolute-error units
14	Mean	0.846	4.70%	hundreds of sheets
	Variance	3.192	19.95%	$(\text{hundreds of sheets})^2$
15*	Mean	1.857	12.38%	days
	Variance	5.143	21.43%	$(\text{days})^2$
16*	Mean	0.700	8.75%	kg
	Variance	5.400	77.14%	$(\text{kg})^2$

